



LIGHT-DEPENDENT RESISTOR (LDR)





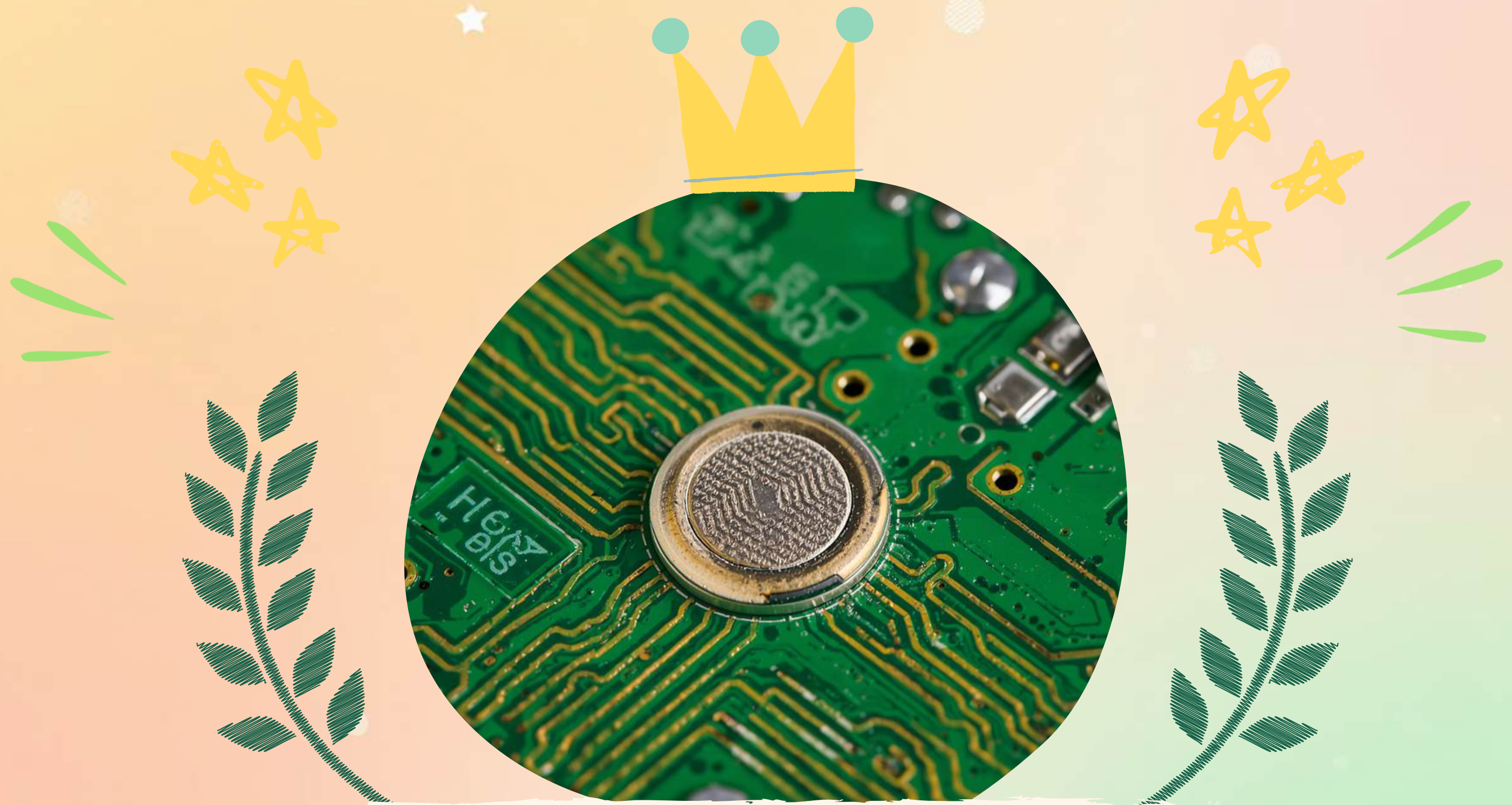
HOW DO STREETLIGHTS KNOW
WHEN TO GLOW?

A SENSOR IS LIKE MINI-EYES FOR ELECTRONICS!

SENSOR



It helps things 'see' the world around them!
Example: Delhi Metro's automatic doors
open by sensing people nearby —
just like a pair of invisible eyes
watching and reacting!



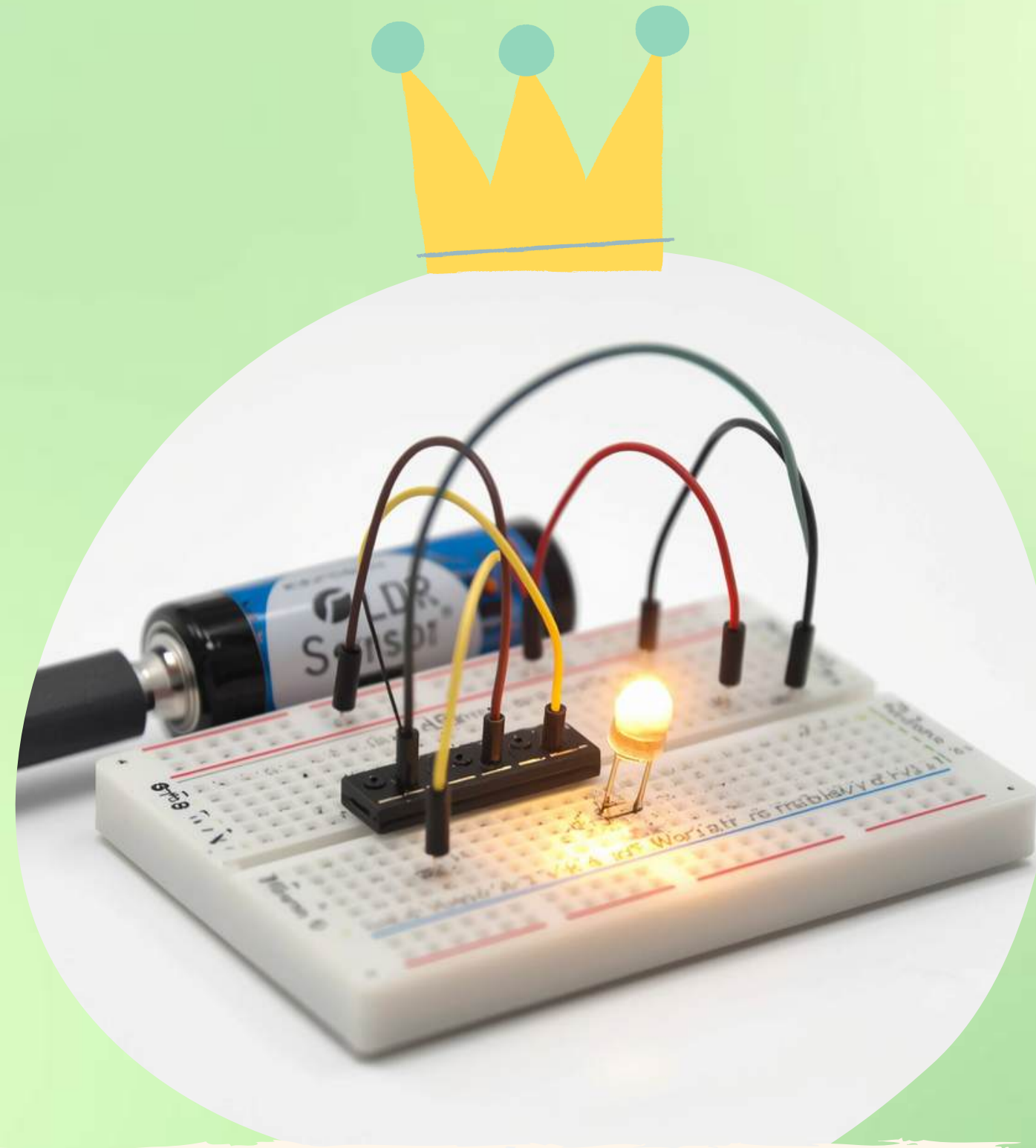
MEET THE LDR — DARKNESS
DETECTIVE!

HOW DOES LDR WORK? — THE WAVY LINE STORY

ALERT!



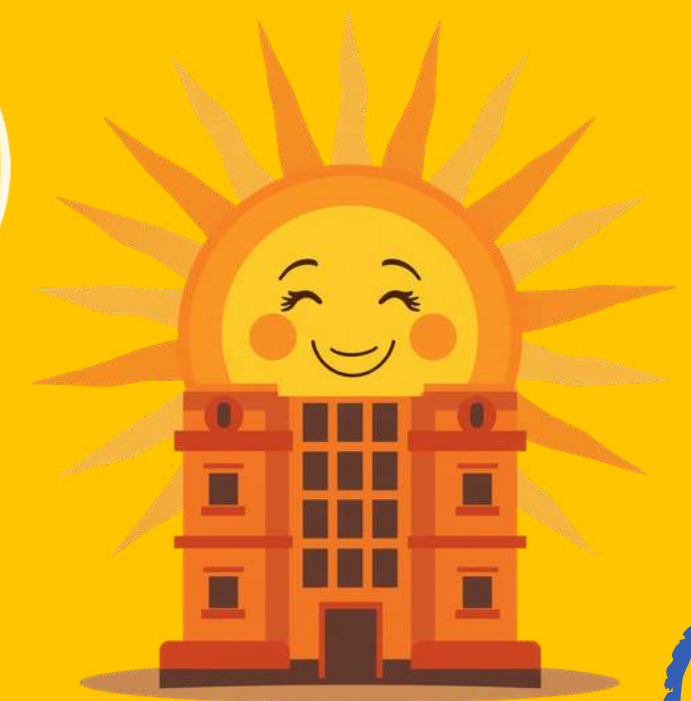
When bright light waves hit LDR, it relaxes. 🧐☀️
When darkness covers it, LDR becomes alert! 🌙
Light = Low resistance. Dark = High resistance!
That's how LDR decides when to switch lights ON!



THE AUTOMATIC NIGHT LIGHT CIRCUIT

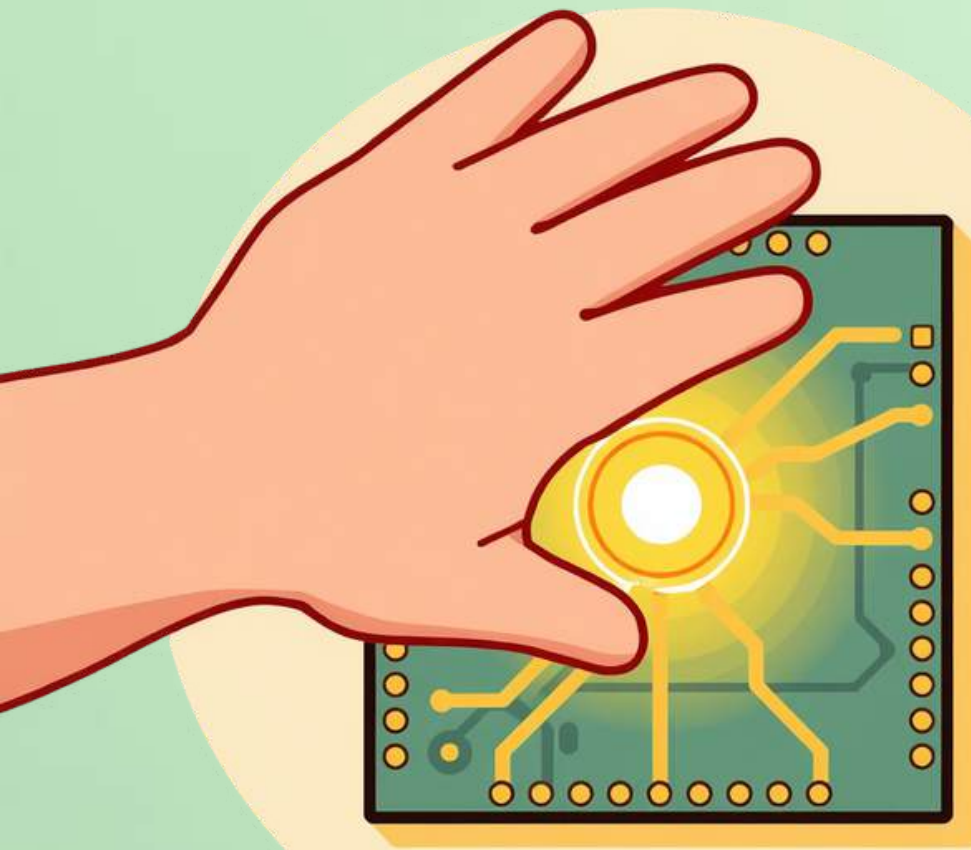
WATCH THE LED LIGHT TURN ON WHEN THE

SUN SETS!

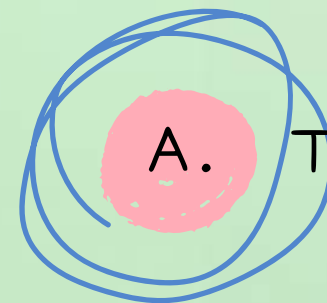


In our virtual circuit, move the sun down
and watch the LED glow bright!
When the sun rises, the LED turns OFF.
The LDR sees the change — just like
magic!

INTERACTIVE CHECKPOINT 2



IF WE COVER THE LDR SENSOR
WITH OUR HAND, WHAT HAPPENS
TO THE LED LIGHT?



Turns On



Turns Off



Blinks



Nothing Happens

REAL-WORLD CONNECTIONS IN INDIA



SOLAR
STREETLIGHTS



SMART CITY
LIGHTS



RAILWAY
COACH LIGHTS



DIWALI
NIGHT BULBS



SAVE POWER!



QUIZ TIME!

WHAT DOES A SENSOR DO?

A. MINI-EYES FOR ELECTRONICS

B. EARS FOR LISTENING

C. HANDS FOR TOUCHING

D. NOSE FOR SMELLING



QUIZ TIME! — QUESTION 2

WHEN DOES THE LDR GET
SUPER ACTIVE?

A. BRIGHT LIGHT

B. DARKNESS

C. RAIN

D. WIND





QUIZ TIME!



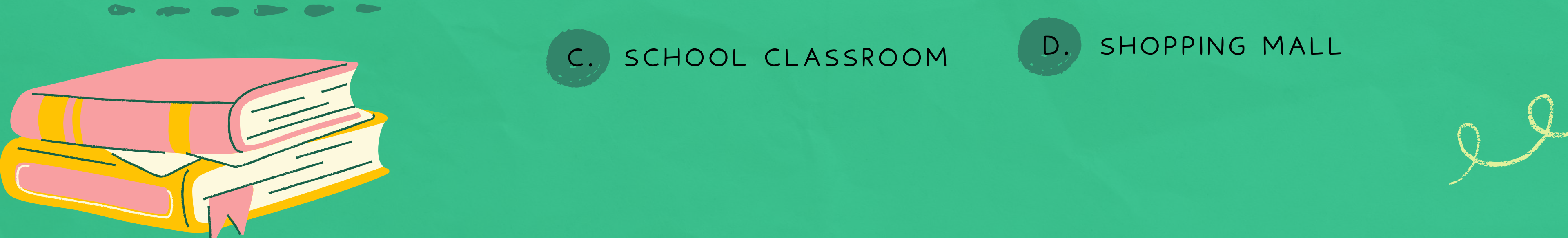
Q3: NAME ONE PLACE IN INDIA WHERE
AUTOMATIC LIGHTS USE LDR.

A. METRO STATION

B. VILLAGE STREETLIGHT

C. SCHOOL CLASSROOM

D. SHOPPING MALL



SUMMARY

WHAT WE
DISCOVERED
TODAY!



SENSORS ARE MINI-EYES FOR
ELECTRONICS!

LDR IS THE DARKNESS DETECTIVE — ON
IN DARK, OFF IN LIGHT!

AUTOMATIC LIGHTS SAVE
POWER EVERY DAY!

COVER LDR = LIGHT ON. BRIGHT
LIGHT = LIGHT OFF!



Thank you for learning with us!

**ANY
QUESTIONS?**

